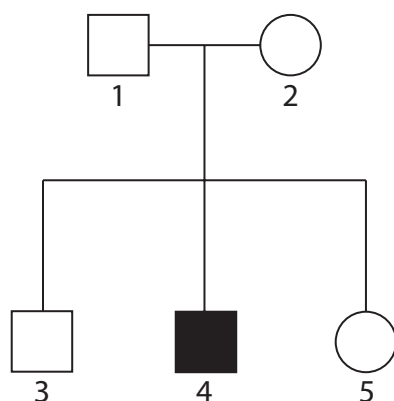


- 3 Charcot-Marie-Tooth (CMT) is an inherited disease that results in the gradual loss of function and sensation in the hands, arms, legs and feet.

Over 100 different gene mutations have been identified that cause CMT.

- (a) The pedigree diagram shows the inheritance of one form of CMT in part of a family.

The allele for CMT can be represented by the letter c in this **recessive** pattern of inheritance.



Key

Unaffected female

Unaffected male

Female with CMT

Male with CMT

- (i) Which row of the table shows the genotype and phenotype for individual 1?

(1)

	genotype	phenotype
☐ A	CC	unaffected homozygous
☐ B	Cc	unaffected heterozygous
☐ C	unaffected heterozygous	CC
☐ D	unaffected homozygous	Cc

(ii) Which allele combinations could individual 3 have?

(1)

- ☐ **A** CC only
- ☐ **B** cc only
- ☐ **C** CC or cc
- ☐ **D** CC or Cc

(iii) What is the probability that a fourth child would be a female with CMT?

(1)

- ☐ **A** 1 in 2
- ☐ **B** 1 in 4
- ☐ **C** 1 in 8
- ☐ **D** 1 in 16

(b) Other forms of CMT show a dominant pattern of inheritance.

(i) In the USA, 5% of people with CMT have one of these dominant forms of the disease.

In the USA, there are a total of 126 000 people with all forms of CMT.

Calculate the number of people in the USA who have one of these dominant forms of CMT.

(1)

Answer



- (ii) An individual with a dominant form of CMT will have the allele combination TT or Tt.

Draw a genetic diagram to show the possible allele combinations of the children of the following parents:

- parent 1 does not have CMT
- parent 2 has the allele combination Tt

Indicate whether each child will have CMT or be unaffected.

(3)

- (c) Another form of CMT can be inherited as a sex-linked trait on the X chromosome.

Explain why a male with CMT **cannot** pass the disease on to his **sons**.

(2)

(Total for Question 3 = 9 marks)

